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Paleomagnetic evidence for clockwise rotation of the northern arm of Sulawesi, Indonesia

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Paleomagnetic results from the northern arm of Sulawesi show that the arm has been subjected to a clockwise rotation of more than 90° and that its rotational motion began no later than the middle Miocene. The mean direction showing a normal polarity at the Eocene to the early Miocene is $D=98.0°$ and $I=6.9°$. A declination value of $D=50.1°$ obtained from Miocene rocks indicates a transition stage of the rotational motion. The datum from Plio-Pleistocene volcanics is $D=-4.6°$ and $I=-9.3°$. This suggests that the rotational motion terminated before the initiation of volcanic activity during the Plio-Pleistocene.

1. Introduction

Sulawesi is a peculiar K-shaped island situated in the central region of the Indonesian archipelago (Fig. 1). Its typical outline has long drawn the attention of geologists and geophysicists. Published reports on the geology of Sulawesi [1–5] suggest that the island is divided into two geological provinces. In the Eastern Province which consists of the eastern and southeastern arms, the geology is dominated by ophiolites and metamorphic rocks, which are locally overlain by Mesozoic sedimentary rocks. These Mesozoic sedimentary strata are closely comparable with successions in the Banda Arc islands ofTimor and Ceram and with parts of Australia. In the Western Province which consists of the southern and northern arms, a pre-Cretaceous metamorphic basement is overlain by volcanic and clastic successions of Tertiary and Quaternary age and associated plutonic intrusions.

Several hypotheses have been put forth to explain the origin of the K-shaped morphology of Sulawesi from a geological viewpoint [6–10]. A paleomagnetic study offers an independent method for testing these hypotheses. During the course of an expedition to the northern arm of Sulawesi in 1978, samples were taken from the Gorontalo and Menado regions for paleomagnetic study. In this paper, we examine the hypothesis of a clockwise rotation of the northern arm of Sulawesi using the results obtained from a paleomagnetic study.

2. Geology and sampling

In the Gorontalo area (see Fig. 1), the basement consists of Eocene to early Miocene oceanic basalt and minor pelagic sediments of the Tinombo Formation [11]. These basement rocks are overlaid with intermediate and acid calc-alkaline volcanics of Miocene age (Bilungala Volcanics). The Bilungala Volcanics were later intruded by dioritic rocks (Bone Diorite) whose age has been de-

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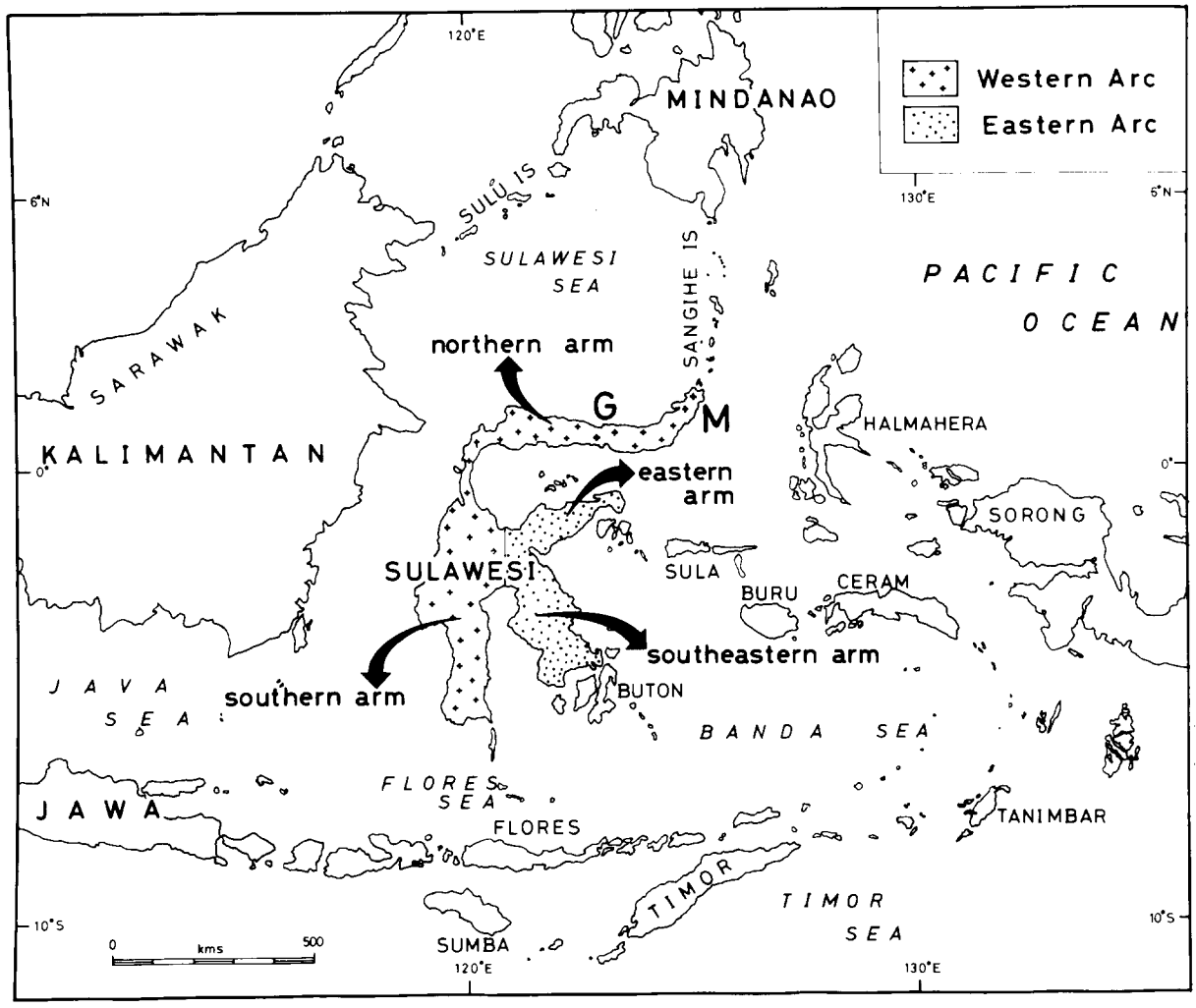


Fig. 1. Map showing the Indonesian archipelago and location of Sulawesi, with position of the Gorontalo and Menado regions. G=Gorontalo region, M=Menado region.

termined as 16 m.y. by K-Ar dating on hornblende [11]. Acid pyroclastic rocks (Pinogu Volcanics) were later produced in the Plio-Pleistocene. The regional geology of the eastern part of the sampling area is shown in Fig. 2. The geological map is that of Lowder and Dow [12].

Rocks of the Tinombo Formation were collected from three sites. An andesite dyke, which intrudes into the Tinombo Formation at the site of IS 120, is labeled IS 120-I in Table 1. Rocks were sampled at three sites from each of the Bilungala and the Pinogu Volcanics. In an area near Menado (see Fig. 1), Tertiary aged pillow lavas were collected at three sites. These sampling locations are listed in Table 1.

3. Paleomagnetism

About ten oriented hand samples were collected at each site. Core specimens, 25 mm in diameter, were drilled in the laboratory. These samples were measured on a spinner magnetometer (Schonstedt, SSM-1A). Preliminary demagnetization tests indi-

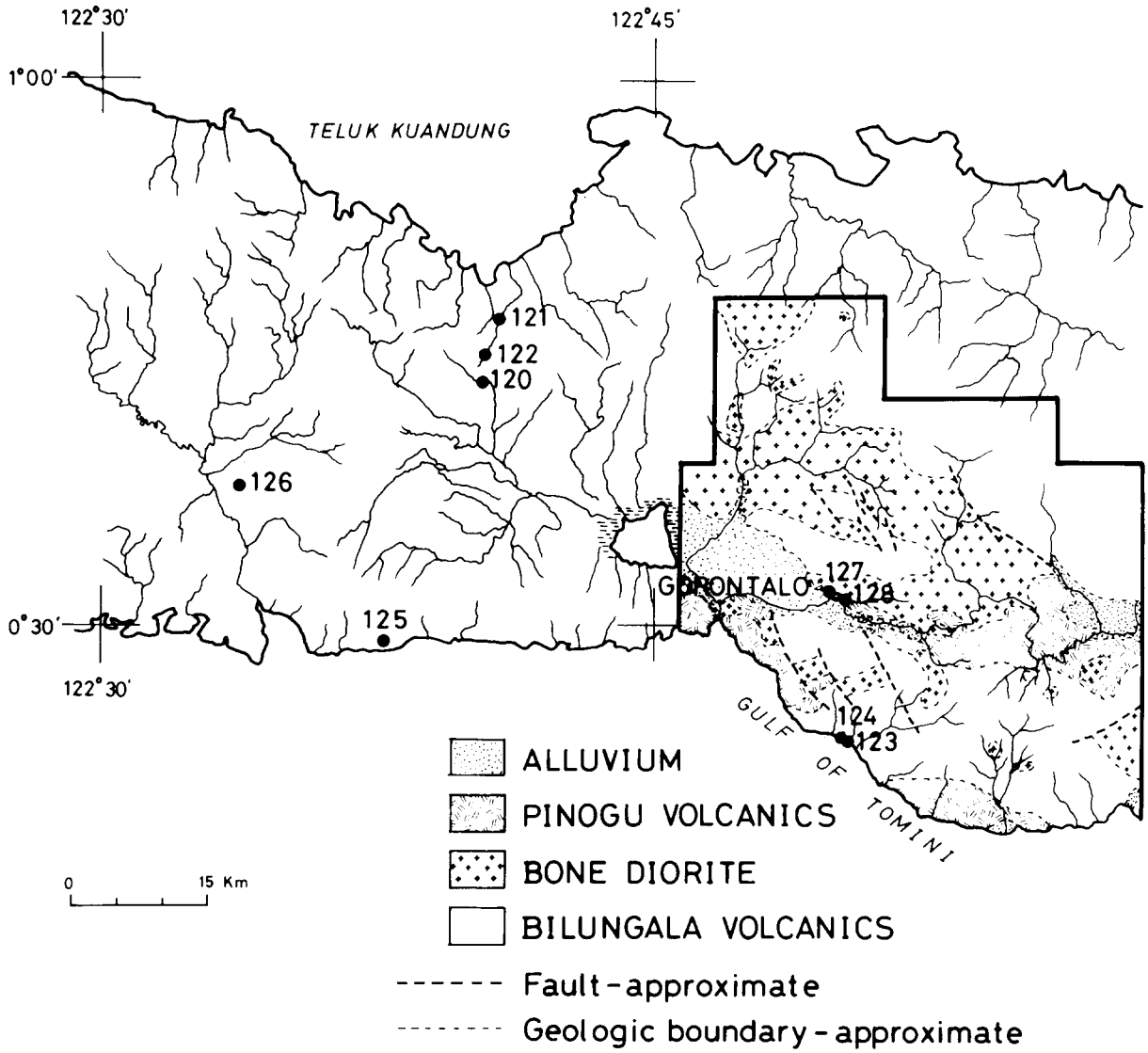


Fig. 2. Map showing sampling locations in the Gorontalo region together with geology about which Lowder and Dow [12] reported only in the area enclosed by the heavy line.

cated that remanent magnetizations of rocks from almost all sites showed different behaviors in response to an alternating field (AF) and thermal demagnetization. All specimens were, therefore, demagnetized stepwise by both methods. The behaviors of natural remanent magnetization (NRM) in response to demagnetizations are described in detail later. Generally, the AF demagnetization technique could not erase the secondary magnetization of rocks. However, the directions of primary magnetization could be obtained by the thermal demagnetization technique. These results are listed in Table 1 and shown in Fig. 3.

3.1. Tinombo Formation

Specimens from IS 120 showed stable primary magnetization, as is indicated by AF and thermal

TABLE 1

Results of paleomagnetic measurement *

Site	Locality	Estimated age	Rock type	N	Demagnetization (th/AF)	D (°)	I (°)	α_{95} (°)	k
Gorontalo region									
Pinogu Volcanics									
IS 128	0°31′N, 123°11′E	Plio-Pleistocene	tuff breccia	10	400°C/100 Oe	−4.6	−9.3	4.4	121.9
Bilungala Volcanics									
IS 121	0°47′N, 122°52′E	Miocene	andesite tuff	7	500°C	49.7	37.2	5.6	115.4
IS 122	0°45′N, 122°51′E		andesite tuff	6	300°C	230.7	51.8	9.6	49.8
Mean						50.1	−7.3		
Tinombo Formation									
IS 120	0°43′N, 122°51′E	Eocene to early Miocene	pyroclastic flow	10	500°C/300 Oe	−85.9	24.3	5.2	88.2
IS 120-1	0°43′N, 122°51′E		andesite tuff	5	500°C	95.4	3.5	26.2	7.8
IS 123	0°23′N, 123°10′E		welded tuff	8	500°C	106.4	42.1	19.2	9.3
Mean						98.0	6.9		
Menado region									
IS 129	1°14′N, 124°28′E	Tertiary	pillow lava	6	500°C/300 Oe	24.0	12.5	13.5	25.4
IS 131	1°07′N, 124°49′E		pillow lava	9	300°C/300 Oe	25.4	37.8	7.6	46.6
Mean						24.6	25.2		

* N = number of specimens, th = thermal demagnetization temperature, AF = demagnetization field, D = declination, I = inclination, α_{95} = radius of 95% confidence circle, k = precision parameter.

demagnetization tests. The directions of magnetization, after 300 Oe AF treatment, were similar to those obtained after 500°C thermal treatment (Fig. 3). Specimens from IS 120-I had stable magnetization in response to AF demagnetization in peak fields up to 300 Oe, whereas the NRM direction moved toward the horizontal east direction in the sequence of the progressive thermal demagnetizations. The mean direction after 500°C thermal treatment was almost anti-parallel to that of IS 120 (Fig. 3).

There was a great shift of the NRM directions of specimens from both IS 123 and IS 124 during the thermal demagnetization (Fig. 4a), even in those specimens that demonstrated high stability in response to AF demagnetization. The directions of NRM from IS 123 once formed a cluster in a shallower and southerly direction after 200–300°C thermal treatment. Subsequently, these moved toward the inclined east direction and formed a cluster again (Fig. 4a). The mean direction of IS 123 after 500°C thermal treatment is, therefore, accepted as the primary component of the NRM, though these directions appear fairly scattered (Fig. 3). Specimens of IS 124 after 400°C thermal treatment have nearly the same NRM directions as specimens of IS 123 after 200–300°C thermal treatment (Fig. 4b,c). It is thought that these directions are originated under the influence of the secondary magnetization produced by a partial thermo-remanent magnetization.

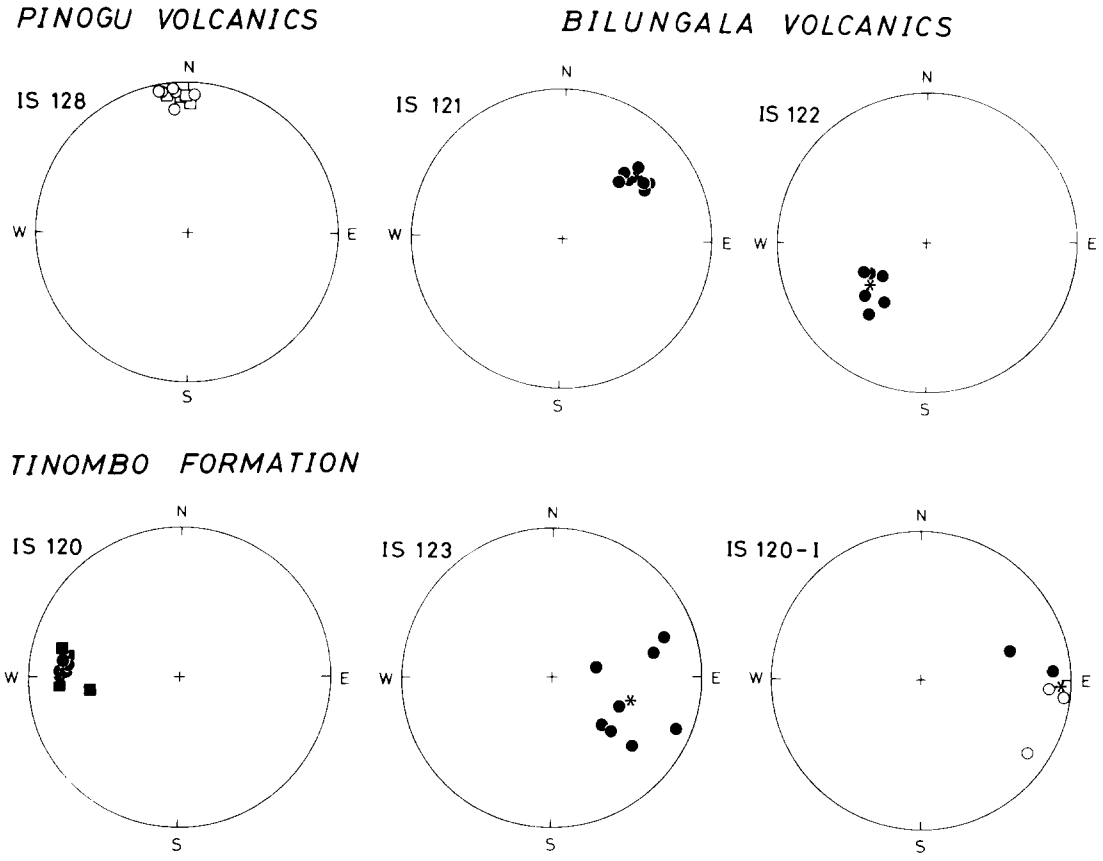


Fig. 3. Equal-area projections showing cleaned directions of magnetization. Remanences after thermal and AF demagnetization are shown with circle and square symbols, respectively. Temperature and field strength of the demagnetization are listed in Table 1. Solid symbols on the lower hemisphere, open symbols on the upper hemisphere. The mean directions except IS 120 and IS 128 are shown with asterisk symbols.

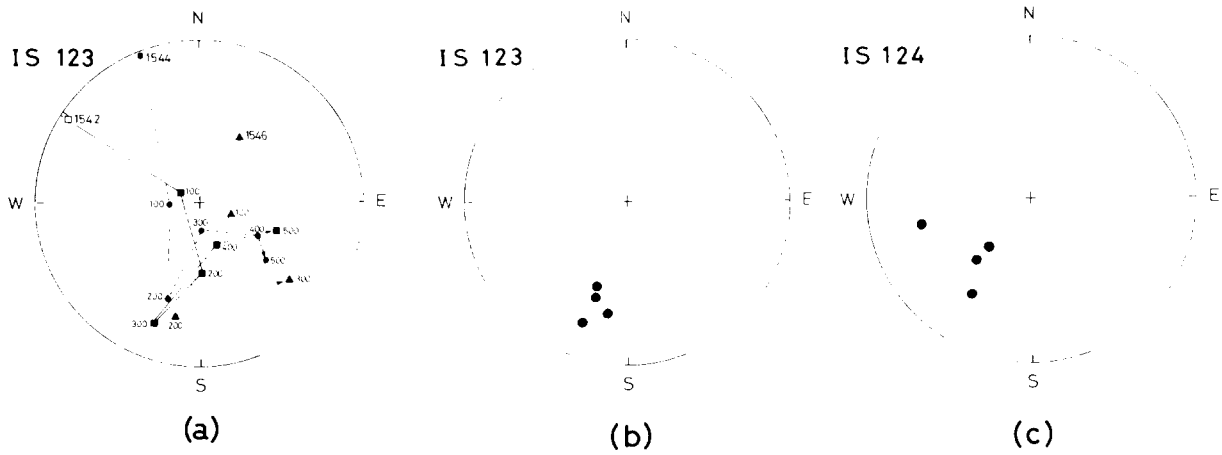


Fig. 4. Equal-area projections showing directions of magnetization: (a) Example of thermal demagnetization of three specimens of IS 123. The directions significantly change in response to demagnetization temperature (numerals, in °C). (b) The directions of remanence (specimens of IS 123) thermally demagnetized at lower temperature (200–300°C). (c) The directions of remanence (specimens of IS 124) after thermal demagnetization at 500°C.

3.2. Bilungala volcanics

A small unstable component of specimens from IS 121 was minimized by an alternating field of only 100 Oe. The NRM directions after 500°C thermal treatment were more tightly clustered than those demagnetized by the AF method (Fig. 3). The unstable component of specimens from IS 122 could not be adequately removed by AF demagnetization in peak fields of 300 Oe. Demagnetization above 300°C succeeded in the removal of these secondary component (Fig. 3).

The NRMs of all specimens from IS 125 (pyroclastic flow) have a horizontal northerly direction that suggests acquisition in the present-day geomagnetic field at the sampling site. The NRM directions became increasingly scattered in sequence of the progressive thermal demagnetization. The precision parameter [13] after 500°C thermal treatment decreased to 1.6. This suggests that the stable remanent component was probably not acquired during deposition.

3.3. Pinogu volcanics

The NRM of specimens from IS 128 was extremely stable. Changes in direction were barely observable after treatment by both AF and Th demagnetization methods (Fig. 3).

Both AF and Th demagnetization induced large changes in the concentration of the NRM directions of specimens from IS 126. The precision parameter k decreased to less than 10 after 150 Oe AF treatment and 300°C thermal treatment. The precision parameter k of NRMs from IS 127 was less than 5 at every demagnetization level. These facts suggest that rocks of both sites (which consists of tuff breccia) did not acquire primarily stable thermo-remanent magnetization during their formation.

4. Significance of results

The reliability of remanent directions as a paleomagnetic record is ascertained through the presence of opposite directions for both of the Tinombo Formation and the Bilungala Volcanics (Fig. 3). Although directions of IS 123 show a fairly steep positive inclination value, the east direction in declination is compatible with the declination obtained from IS 120-I. As the secular variation causes a great inclination change near the equator, the steep positive inclination of IS 123

may be explained by the geomagnetic secular change. Steep inclination values of the Miocene Bilungala Volcanics also may be due to the secular variation and/or due to the tectonic tilting accompanied by Miocene uplift in the northern part of Sulawesi [6].

One of the difficulties in working with rocks near the equator is that the inclination data are usually a reliable index of polarity. However, the polarity can be determined from the declination change as a function of time. It is noted that there are no specimens whose remanent magnetizations are either in a northwest or southeast direction (Fig. 3). This implies that the NRMs of specimens of IS 120-I and IS 123 were acquired in the same geomagnetic polarity field as the NRM of IS 121 and that the NRMs of IS 120 and IS 122 were acquired in the field polarized oppositely to the former. By comparison of the two directions of IS 121 and IS 122 (Fig. 3), specimens of IS 121 are found to have had normal polarities. It is therefore interpreted that the subhorizontal west direction of IS 120 in the Tinombo Formation was acquired during reversed geomagnetic interval.

The mean directions of normal polarity are listed in Table 1 for each geological time of the Gorontalo region and for the Tertiary of the Menado region.

5. Discussion

The inclination data from the Gorontalo region give the paleo-latitude of 4.7°S, 3.7°S and 3.5°N for the Plio-Pleistocene, the Miocene and the Eocene/early Miocene, respectively. This fact indicates that the northern arm of Sulawesi has been situated at its nearly present latitude since the Eocene.

Directions of the declination are drawn in Fig. 5. Data from the Tertiary pillow lavas from the Menado region are shown for the Eocene/early Miocene epoch; declinations for the Menado region in the Plio-Pleistocene are from earlier work [14].

The remanence of specimens magnetized during a normal polarity interval in the Eocene/early Miocene point at present toward the east (Fig. 5). This direction is explained by a clockwise rotation through roughly 90° of the northern arm of Sulawesi during or after the epoch of the Eocene/early Miocene. The 90° rotation is further supported by the declination data of the Miocene epoch which indicates a transition stage. The clockwise rotation of the arm favors the southwest direction of secondary magnetization observed in IS 123 and IS 124. These paleomagnetic data confirm the clockwise rotation suggested by the present geomorphological configuration [6,9]. This clockwise rotation occurred, strictly speaking, after the time of intrusion of the andesite dyke, later than the Eocene to early Miocene, into the site of IS 120, and the rotational motion ceased before volcanic activity renewed in the Plio-Pleistocene.

The Menado region appears not to have undergone a large rotational movement from the geomorphological configuration of Sulawesi. This appearance may be reflected in the declination values of about 25° obtained from the Menado region (Fig. 5). The occurrence of a clockwise rotation of more than 90° is limited to the central part of the northern arm of Sulawesi.

Two possibilities are considered for the clockwise rotational manner of the arm: The pole of the rotation lay either at the western end of the northern arm or at the eastern end. The paleomagnetic data obtained from the Gorontalo region do not allow us to choose between these possibilities. Based on an indication of good correlation between the geology of western Sulawesi with that of Sangihe Island and the Philippines to the north [6], it is inferred that the pole of the rotation lay at the eastern end near Menado. Before the northern arm was subjected to the clockwise rotational motion, Mindanao, Sangihe Island and the northern arm of Sulawesi may have composed a nearly straight north-south-trending island chain. Only the northern arm would rotate and override the Celebes Sea floor by subduction at the North Sulawesi Trench [15]. This speculation may account for the observed data that the rotational angle of the Gorontalo region is much larger than that of the Menado region.

An angular velocity is roughly estimated for the rotation of the northern arm of Sulawesi: The velocities of about 2°/m.y. and about 4.5°/m.y.

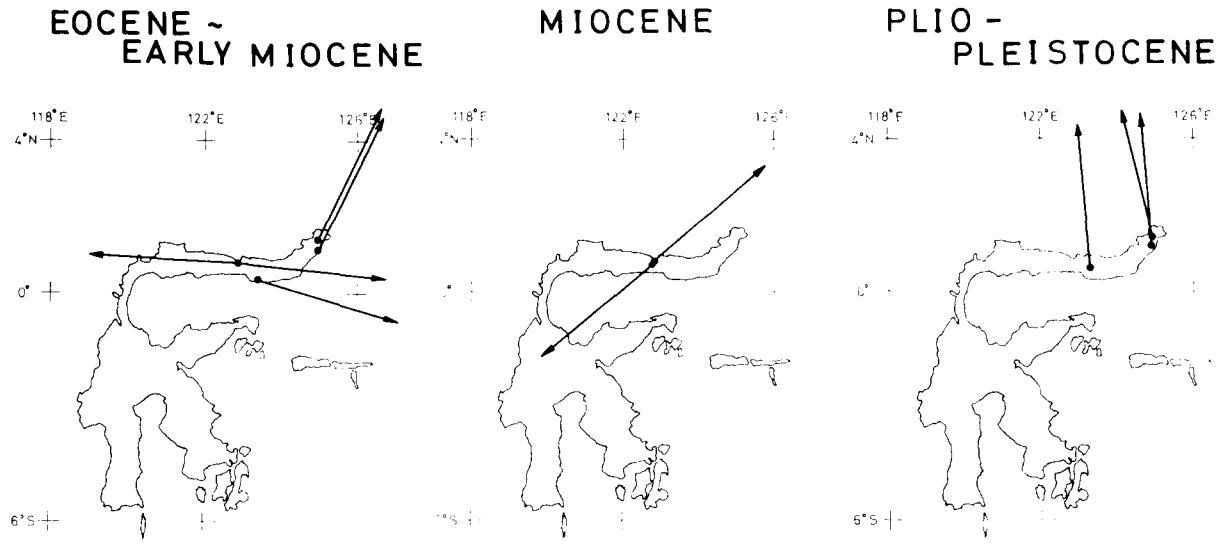


Fig. 5. Declination changes in the Gorontalo and Menado regions since the Eocene to early Miocene.

are obtained by the assumption that the beginning of the rotation is either in the Eocene or the Miocene. These data are compared with available rotational motion from the other tectonic areas. The Italian Peninsula was rotated clockwise through 40° between the late Jurassic and the early Tertiary [16]. A major phase of rotation over about 30° occurred during Turonian, Coniacian, and Santonian time. The angular velocity attains a maximum of 1.5°/m.y. Kawai et al. [17] discussed the bending of the arc of Japan during the Cretaceous. Their postulated angular velocity of bending is about 2.4°/m.y. Paleomagnetic data from the Oregon Coast Range [18] indicate a clockwise rotation of at least 75° with a velocity of about 2.1°/m.y. between the Eocene and the middle Miocene. These suggest that although the estimated rotation of Sulawesi may be a little faster than the rotation observed in other areas, it is not an unreasonable rate for a tectonic area.

6. Conclusion

The paleomagnetic study supports a clockwise rotational motion of the northern arm of Sulawesi. This 90° rotation occurred after the Eocene/early Miocene period. The pole of rotation was located at the eastern end of the northern arm, near Menado. The rotational motion ceased before the volcanic activity was renewed at the Plio-Pleistocene.

This paleomagnetic study and other geological works [6,7,10,15] allow us to reconstruct the evolution of Sulawesi as follows: The Western Province of Sulawesi belonged to South Asia, which was situated far away from the Australian continent, during the Permian and Mesozoic. Since the Cretaceous, Eastern Sulawesi, Sulu, Buru and Ceram (which were at that time part of the northern margin of Australia) were moving westward toward the South Asian continent associated with the northward drift of Australia. This motion resulted in the collision of eastern Sulawesi with western Sulawesi. The northern arm of Sulawesi, which had composed a nearly straight north-south-trending island chain with Mindanao and Sangihe Island, was rotated clockwise more than 90° due to the collision. The westward motion caused the occurrence of the large ophiolite belts east of western Sulawesi as well as the clockwise rotation of the northern arm.

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